FULL APLLICATION AND DEVELOPMENT

#SKILL - 10 ➔ React State Management using useState Object

Prerequisites:

- **Basic Idea of JavaScript**
- **Knowledge of React fundamentals**

An online academic portal wants to maintain a list of **students** where the user can **add new students**, **display the list**, and **delete a student** instantly without refreshing the page. The application should use **React Hooks** to manage data at the component level so that the UI updates automatically whenever the state changes.

Your task is to implement a React component using **useState** to store an **object array**, add new items, display them, and delete them dynamically.

1. Create a new React component named **StudentManager**.
2. Inside the component, create an initial **students array** with at least 5 objects containing:
   a. **id**
   b. **name**
   c. **course**
3. Use **useState** to store:
   a. **students** (array of student objects)
   b. **newStudent** (object with fields like id, name, course)
4. Create **input boxes** to accept id, name, and course.
   i. Update the **newStudent object** using **onChange** events.
5. Create an **Add Student** button.
   i. On click, add the **newStudent object** to the students array.
   ii. After adding, **clear the input fields**.
6. Display the list of students below using the **students state**.
7. Each student row should have a **Delete button**.
   i. On clicking delete, remove the record from the students array.
   ii. The UI should update immediately using **setStudents**.
8. If the list becomes empty, display the message:
   "No students available"
9. Add **basic CSS styling** for a clean UI (margin, padding, table style, buttons).
10. **Push the React project to GitHub.**

Create Project in VS code Using Vite

Open terminal and run:

npm create vite@latest student-manager

Select:

Framework : React
Variant : JavaScript

In bash run:

cd student-manager
npm install
npm run dev

Project Structure

```
student-manager
│
├── src
│   ├── components
│   │    ├── StudentManager.jsx
│   │    └── StudentManager.css
│   │
│   ├── App.jsx
│   └── main.jsx
```

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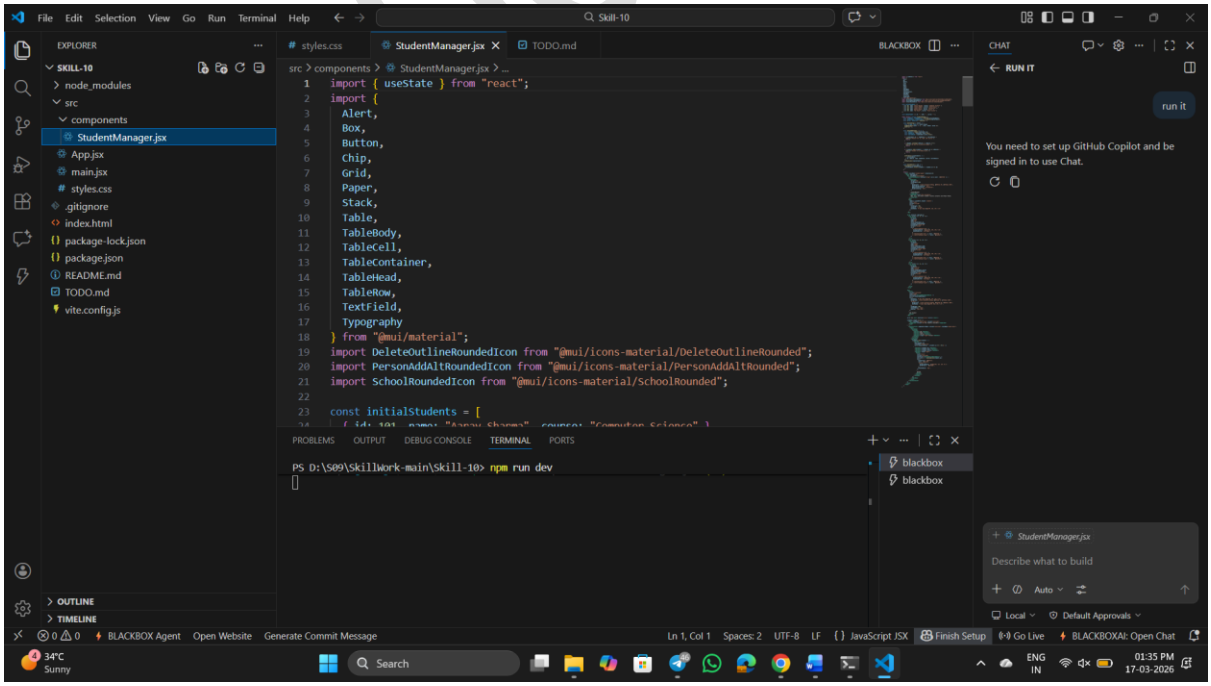
StudentManager.jsx

```
import React,{useState} from "react";
import "./StudentManager.css";

function StudentManager(){
const initialStudents=[
{id:2400031000,name:"Rahul",course:"CSE"},
{id:2400031001,name:"Sneha",course:"AI&DS"},
{id:2400031002,name:"Arjun",course:"IT"},
{id:2400031003,name:"Priya",course:"CS"},
{id:2400031004,name:"Kiran",course:"CSE"}
];

const [students,setStudents]=useState(initialStudents);
```

```
const [newStudent,setNewStudent]=useState({
id:"",
name:"",
course:""
});

const handleChange=(e)=>{
const {name,value}=e.target;
setNewStudent({...newStudent,[name]:value});
};

const addStudent=()=>{
setStudents([...students,newStudent]);
setNewStudent({id:"",name:"",course:""});
};

const deleteStudent=(id)=>{
const updated=students.filter((student)=>student.id!==id);
setStudents(updated);
};

return(

{students.length===0?(

export default StudentManager;
```



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StudentManager.css

```
body {
margin: 0;
height: 100vh;
display: flex;
justify-content: center;
align-items: center;
background: linear-gradient(135deg, #eef2f7, #d9e4f5);
font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;}
.container {
width: 650px;
background: #ffffff;
padding: 30px;
border-radius: 12px;
box-shadow: 0 10px 25px rgba(0,0,0,0.08);
text-align: center;}
.container h2 {
margin-bottom: 20px;
color: #2c3e50;}
.form {
display: flex;
justify-content: center;
flex-wrap: wrap;
gap: 10px;
margin-bottom: 25px;}
input {
padding: 10px;
width: 150px;
```

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```
border-radius: 6px;
border: 1px solid #dcdcdc;
font-size: 14px;}
input:focus {
outline: none;
border-color: #4a90e2;}
button {
padding: 10px 16px;
border: none;
border-radius: 6px;
background: #4a90e2;
color: white;
font-weight: 600;
cursor: pointer;}
button:hover {
background: #357ab8;}
.delete {
background: #e74c3c;}
.delete:hover {
background: #c0392b;}
table {
width: 100%;
border-collapse: collapse;}
th {
background: #4a90e2;
color: white;
padding: 10px;}
td {
```

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padding: 10px;

border-bottom: 1px solid #eee;}

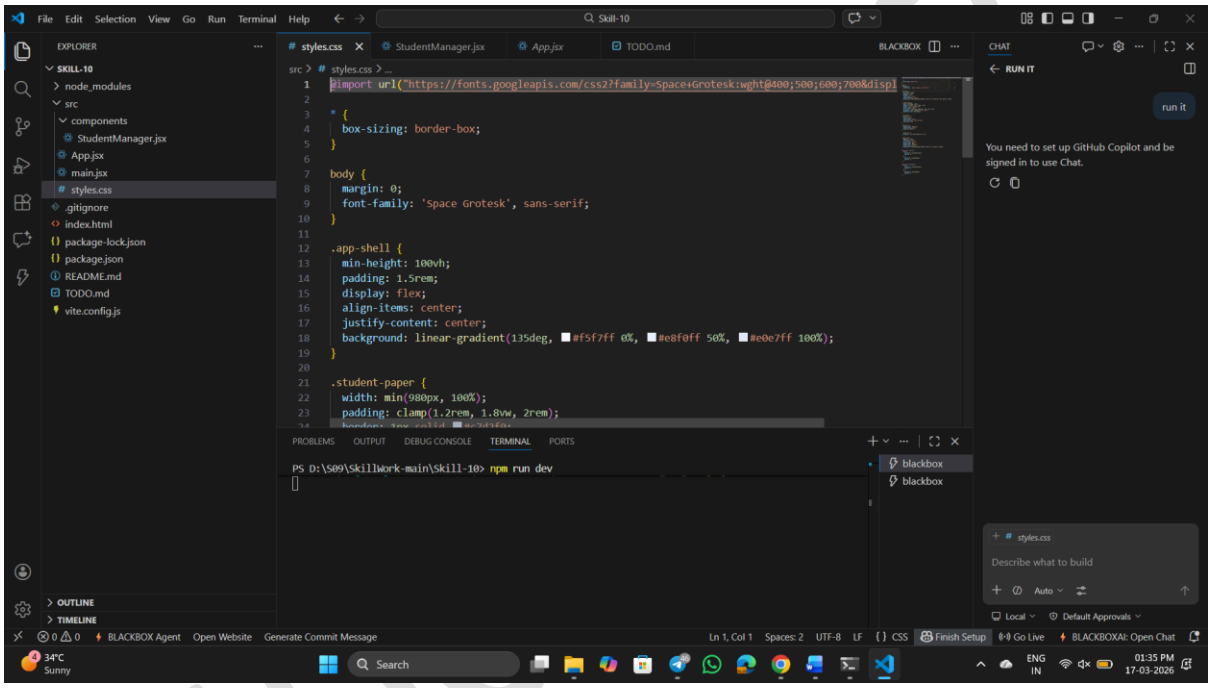
tr:nth-child(even) {

background: #f7f9fc;}

.empty {

margin-top: 15px;

color: #888;}



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App.jsx

import StudentManager from "./components/StudentManager";

function App() {

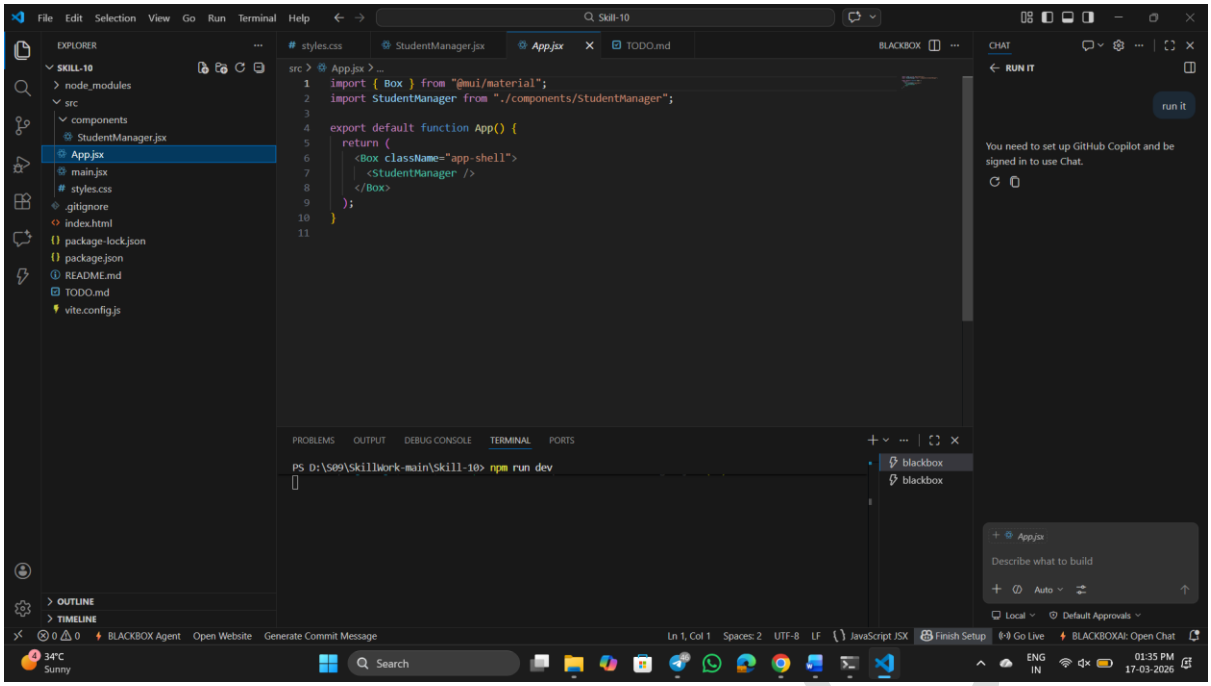
return (

<>

<StudentManager />

</>); }

export default App;



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Run the Application

1. Open terminal in project folder: npm run dev
2. Open browser: http://localhost:5173

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Output:

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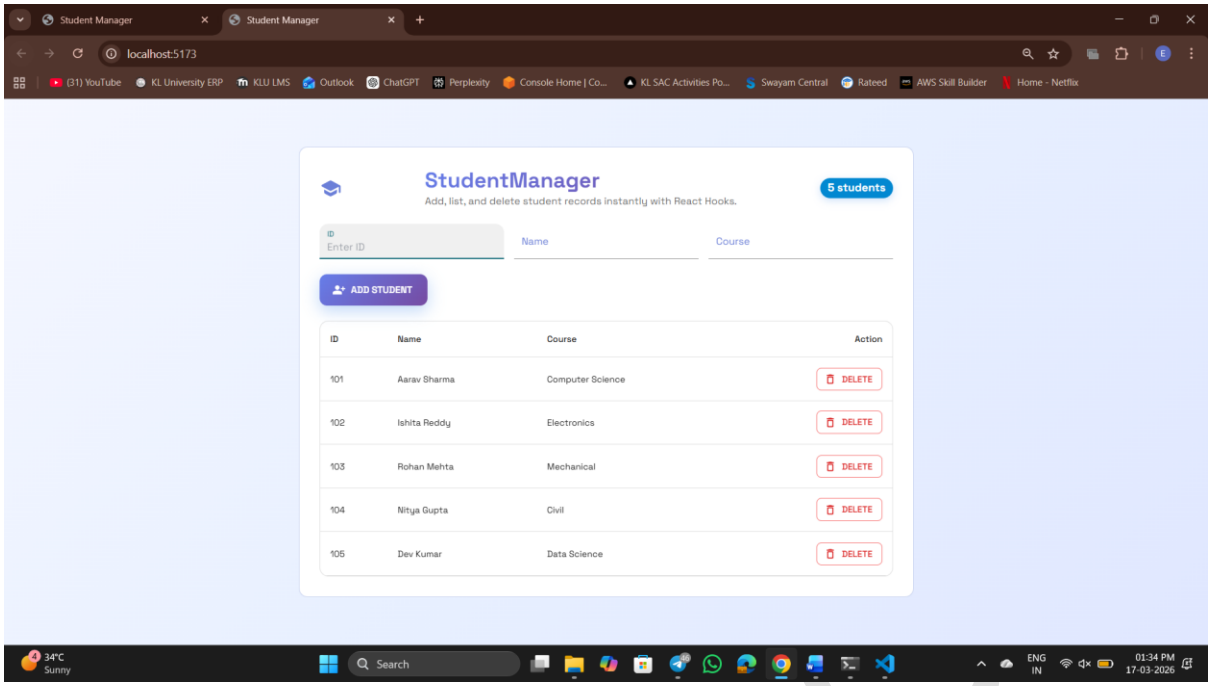
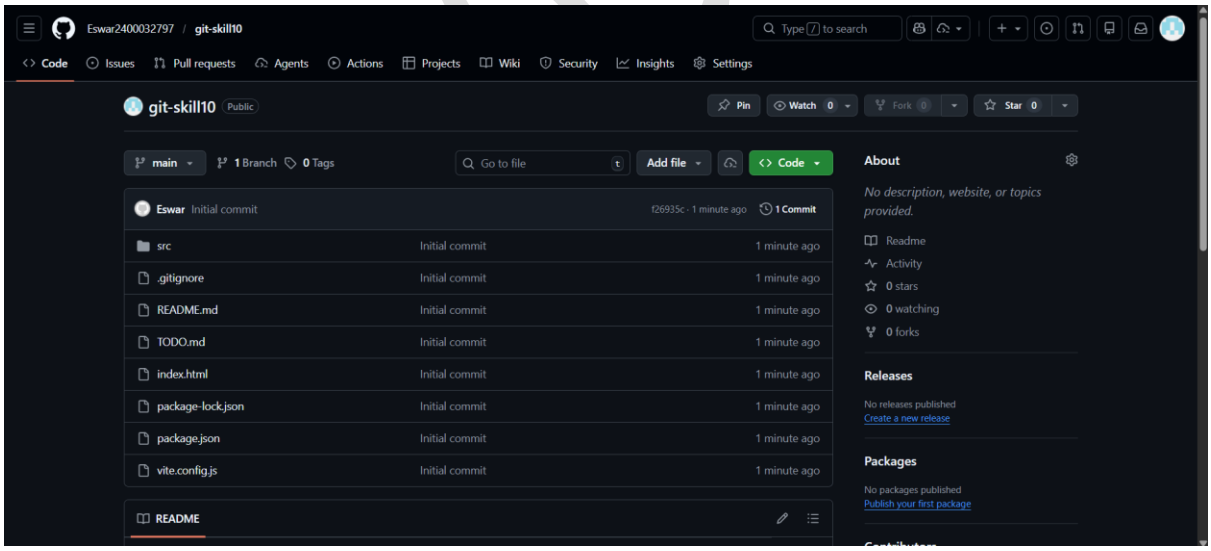


GITHUB OUTPUTS:



GITHUB REPO LINK:

https://github.com/Eswar2400032797/git-skill10.git